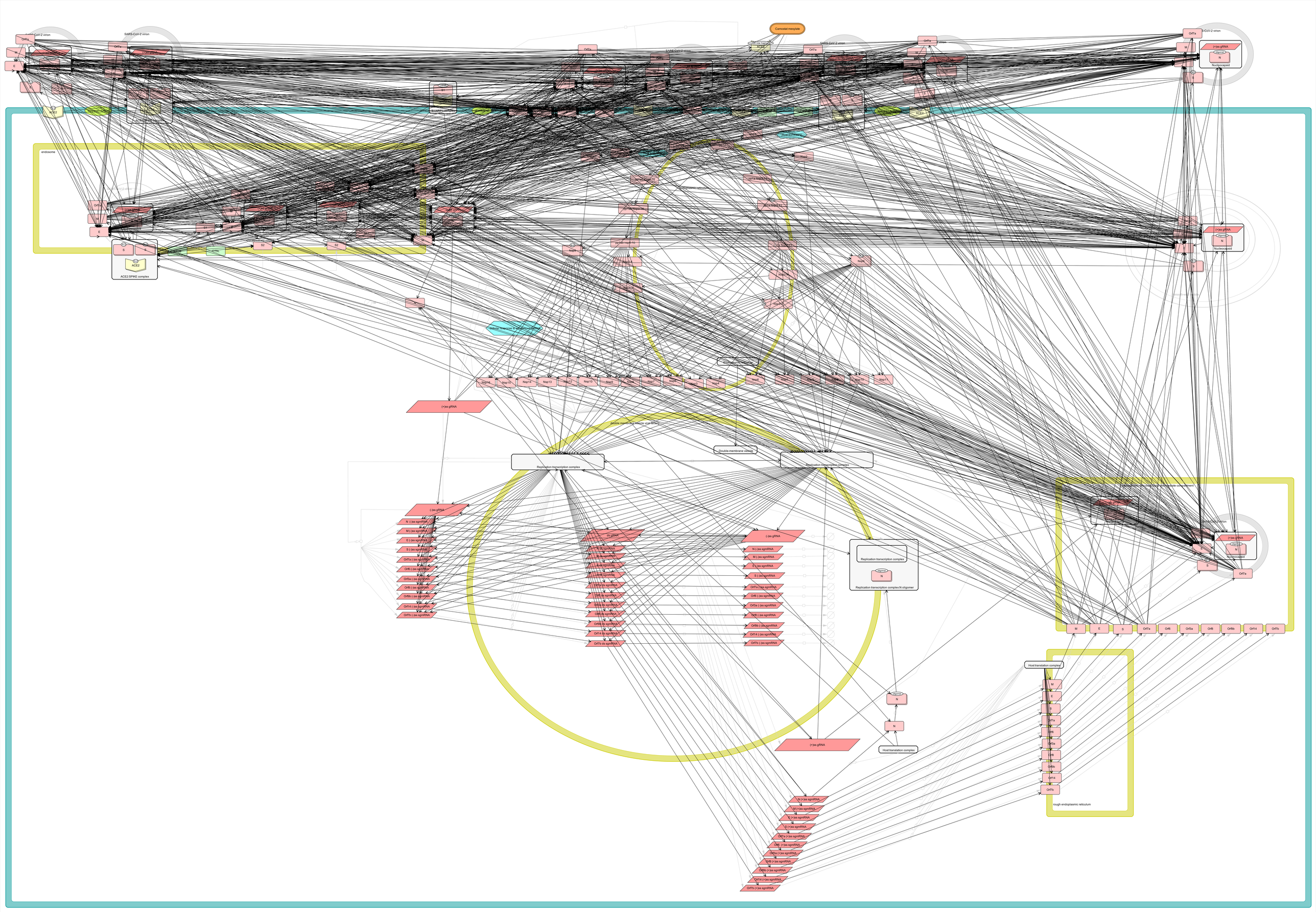

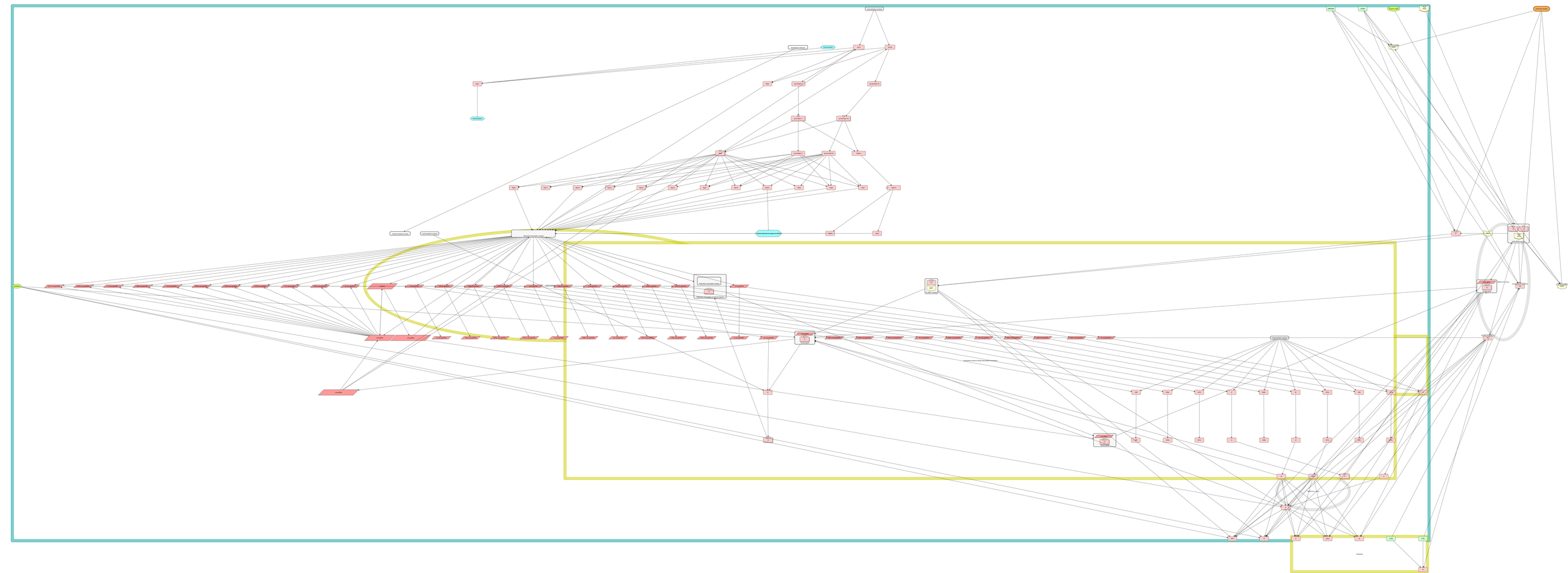

SARS-CoV-2 virion
Orf7a
M
E
S
N:3
(+)ss gRNA
oligomer
N
Nucleocapsid
ACE2
N:2
Heparan sulfate
ACE2:SPIKE complex
epithelial cell
CendR
S1
NRP1
S1:NRP1 complex
Camostat mesylate
100 Kda fragment
ACE2
FURIN
TMPRSS2
Host translation
endosome
S1
S2
exposed HR1:HR2 FP
6-HB conformation
CTSB
CTSL
endoplasmic reticulum
pp1ab Nsp3-16
pp1a Nsp3-11
pp1ab nsp6-16
pp1a Nsp6-11
Nsp2
Nsp5
Nsp3-4
cellular response to exogenous dsRNA
Endoplasmic reticulum
Nsp16
Nsp15
Nsp14
Nsp13
Nsp12
Nsp10
Nsp9
Nsp8
Nsp7
Nsp6
Nsp3
Nsp4
Nsp11
double membrane vesicle viral factory
Double-membrane vesicle
Replication transcription complex
(-)ss gRNA
N (-)ss sgmRNA
M (-)ss sgmRNA
E (-)ss sgmRNA
S (-)ss sgmRNA
Orf7a (-)ss sgmRNA
Orf6 (-)ss sgmRNA
Orf3a (-)ss sgmRNA
Orf8 (-)ss sgmRNA
Orf9b (-)ss sgmRNA
Orf14 (-)ss sgmRNA
Orf7b (-)ss sgmRNA
ds gRNA
N ds sgmRNA
M ds sgmRNA
E ds sgmRNA
S ds sgmRNA
Orf7a ds sgmRNA
Orf6 ds sgmRNA
Orf3a ds sgmRNA
Orf8 ds sgmRNA
Orf9b ds sgmRNA
Orf14 ds sgmRNA
Orf7b ds sgmRNA
Replication transcription complex:N oligomer
N (+)ss sgmRNA
M (+)ss sgmRNA
E (+)ss sgmRNA
S (+)ss sgmRNA
Orf7a (+)ss sgmRNA
Orf6 (+)ss sgmRNA
Orf3a (+)ss sgmRNA
Orf8 (+)ss sgmRNA
Orf9b (+)ss sgmRNA
Orf14 (+)ss sgmRNA
Orf7b (+)ss sgmRNA
Host translation complex
endoplasmic reticulum-Golgi intermediate compartment
Orf6
Orf3a
Orf8
Orf9b
Orf14
Orf7b
rough endoplasmic reticulum
smooth-wall vesicle

Host translation complex
Endoplasmic reticulum
Host translation
pp1a
pp1ab
Nsp1
Host translation
Nsp2
pp1a Nsp3-11
pp1ab Nsp3-16
pp1a Nsp3-11
pp1ab Nsp3-16
Nsp5
pp1a Nsp6-11
pp1ab nsp6-16
Nsp3-4
Nsp8
Nsp11
Nsp16
Nsp12
Nsp14
Nsp13
Nsp9
Nsp10
Nsp15
Nsp8
Nsp6
Nsp7
Nsp3-4
Double-membrane vesicle
Host translation complex
Replication transcription complex
cellular response to exogenous dsRNA
Nsp4
Nsp3
TMPRSS2
FURIN
Heparan sulfate
ACE2
Camostat mesylate
15 Kda fragment
ACE2
CendR
S1
NRP1
S
S
ACE2
ACE2:SPIKE complex
(+)ss gRNA
oligomer
N
Nucleocapsid
SARS-CoV-2 virion
exposed HR1-HR2-FP
S2
100 Kda fragment
ACE2
cholesterol
Orf14 (-)ss sgmRNA
Orf7a (-)ss sgmRNA
E (-)ss sgmRNA
Orf6 (-)ss sgmRNA
S (-)ss sgmRNA
Orf3a (-)ss sgmRNA
Orf9b (-)ss sgmRNA
Orf8 (-)ss sgmRNA
N (-)ss sgmRNA
Orf7b (-)ss sgmRNA
M (-)ss sgmRNA
ds gRNA
N ds sgmRNA
Orf6 ds sgmRNA
Orf3a ds sgmRNA
Orf14 ds sgmRNA
S ds sgmRNA
double membrane vesicle
Orf9b ds sgmRNA
M ds sgmRNA
Orf7a ds sgmRNA
Orf8 ds sgmRNA
Orf7b ds sgmRNA
Replication transcription complex
oligomer
N
Replication transcription complex:N oligomer
E ds sgmRNA
CendR
S1
NRP1
S1:NRP1 complex
(-)ss gRNA
(+)ss gRNA
N (-)ss sgmRNA
Orf6 (-)ss sgmRNA
Orf3a (-)ss sgmRNA
Orf14 (-)ss sgmRNA
S (-)ss sgmRNA
Orf9b (-)ss sgmRNA
M (-)ss sgmRNA
Orf7a (-)ss sgmRNA
Orf8 (-)ss sgmRNA
Orf7b (-)ss sgmRNA
E (-)ss sgmRNA
N (+)ss sgmRNA
(+)ss gRNA
oligomer
N
Nucleocapsid
Orf6 (+)ss sgmRNA
Orf3a (+)ss sgmRNA
Orf14 (+)ss sgmRNA
S (+)ss sgmRNA
Orf9b (+)ss sgmRNA
M (+)ss sgmRNA
Orf7a (+)ss sgmRNA
Orf8 (+)ss sgmRNA
Orf7b (+)ss sgmRNA
E (+)ss sgmRNA
Host translation complex
6-HB conformation
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endoplasmic reticulum-Golgi intermediate compartment
(+)ss gRNA
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oligomer
N
Orf6
Orf3a
Orf14
S
Orf9b
M
Orf7a
Orf8
Orf7b
E
(+)ss gRNA
oligomer
N
Nucleocapsid
Orf6
Orf3a
Orf14
S
Orf9b
M
Orf7a
Orf8
Orf7b
M
Orf7a
S
E
SARS-CoV-2 virion
E
Orf7a
M
E
Orf7a
M
CTSB
CTSL
endosome
S1